

QUIZ 3 SET D – SOLUTION
DURATION: 25 MINUTES

SUMMER SEMESTER, 2024-2025
FULL MARKS: 20

Programmable calculators are not allowed.

ID:

- 5
(CO5)
(PO3)

$$I_3 > I_0 > I_6 > I_2 > I_5 > I_7 > I_4 > I_1$$

- Complete the truth table based on this priority.

[illegible]

Here, x denotes a don't-care input. The output is the ordinary three-bit binary code of the highest-priority asserted input.

Inputs (in decreasing priority)								Outputs			
I_3	I_0	I_6	I_2	I_5	I_7	I_4	I_1	O_2	O_1	O_0	V
1	x	x	x	x	x	x	x	0	1	1	1
0	1	x	x	x	x	x	x	0	0	0	1
0	0	1	x	x	x	x	x	1	1	0	1
0	0	0	1	x	x	x	x	0	1	0	1
0	0	0	0	1	x	x	x	1	0	1	1
0	0	0	0	0	1	x	x	1	1	1	1
0	0	0	0	0	0	1	x	1	0	0	1
0	0	0	0	0	0	0	1	0	0	1	1
0	0	0	0	0	0	0	0	0	0	0	0

Rubrics:

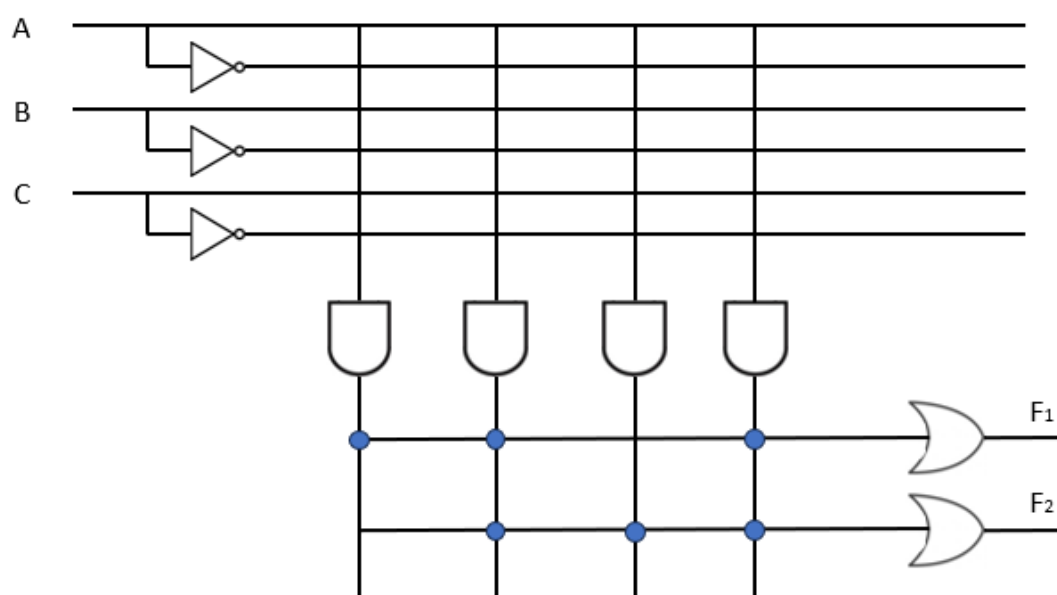
- 1 mark for arranging and labeling the inputs consistently with the stated priority.
- 3 marks for the correct input conditions and encoded outputs for the eight asserted-input cases.
- 1 mark for the valid output, including the all-zero input case.

2. In the following PLD, only the AND array is programmable; the OR array is fixed. The PLD has 3 inputs, A , B , and C , and 2 outputs, F_1 and F_2 . The functions are defined as follows:

$$F_1(A, B, C) = \sum m(1, 2) + d(3), \quad F_2(A, B, C) = \sum m(1, 2, 7) + d(3).$$

Make the appropriate connections to implement F_1 and F_2 using the structure shown below. Note that:

- No structural changes are allowed in the PLD.
- You may not change the positions of the inputs or outputs.
- Mark programmable connections with $\times$. A $\odot$ denotes a fixed connection. Intersecting wires are not connected unless one of these symbols is shown at the junction.
- Simplify the functions as needed, and then implement F_1 and F_2 in the given PLD while following the rules above.



Solution:

Using the don't-care terms where advantageous, the simplified expressions are

$$F_1 = A'B + A'C \quad \text{and} \quad F_2 = A'B + A'C + BC.$$

Thus, the three distinct product terms to be formed in the programmable AND array are $(A'B, A'C, BC)$.

Connect the two product terms appearing in F_1 to its fixed OR gate and all three product terms appearing in F_2 to its fixed OR gate. The programmable junctions may be annotated on the supplied figure.

Rubrics:

- 1 mark for correctly simplifying F_1 .
- 1.5 marks for correctly simplifying F_2 .
- 1.5 marks for realizing the three correct product terms in the programmable AND array.
- 1 mark for assigning the product terms to the correct fixed OR gates.

3. In the following circuit, the inputs A , B , and C are n -bit numbers, and the output F is also an n -bit number.

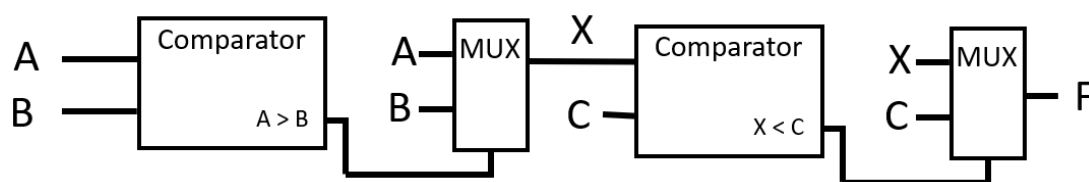
The comparator circuit compares two n -bit numbers M and N and produces a comparison status indicating whether

$$M > N, \quad M < N, \quad \text{or} \quad M = N.$$

Each comparator output is HIGH when its corresponding condition is satisfied. For example, $M > N$ is 1 when M is greater than N , and 0 otherwise.

The MUX is an n -bit multiplexer that selects one of its two inputs based on the selection bit. If the MUX receives M and N as inputs and the selection line is 1, it selects the first input, M .

Based on the given circuit, determine whether F represents the **largest, smallest, median (middle) value**, or something else among the three input numbers A , B , and C . Justify your answer with a suitable example.



Solution:

The first comparator and MUX produce $X = \max(A, B)$. The second stage therefore produces

$$F = \min(X, C) = \min(\max(A, B), C).$$

For example, with $A = 2, B = 7, C = 1, X = 7$ and $F = 1$, the smallest value. With $A = 2, B = 7, C = 4$, however, $F = 4$, the median value.

Therefore, the circuit does *not* always produce the largest, smallest, or median of the three inputs; its result depends on the position of C relative to A and B .

Rubrics:

- 2 marks for correctly identifying the first-stage output X .
- 2 marks for correctly identifying the final expression for F .
- 2 marks for the correct interpretation supported by a valid example (or equivalent case analysis).

4. Answer the following questions.

4 × 1
(CO3)
(PO2)

Solution:

a) What type of PLD is shown in Question 2?

Answer: PAL (programmable AND array and fixed OR array).

b) How many 3×8 decoders are needed to implement an 8×256 decoder?

Answer: $\frac{256}{8} = 32$ decoders. If additional decoders are used to generate the enable signals in a fully cascaded realization, 37 is an alternate answer and will also be accepted as correct.

c) Suppose a 6-bit number is multiplied by a 9-bit number. How many adders are required? (The adders may have any number of bits.)

Answer: $6 + 9 - 2 = 13$ adders.

d) In a BCD adder, would an error occur if 6 were added to the preliminary sum only when the addition produced a fifth-bit carry? Explain in one sentence.

Answer: Yes. A preliminary sum from 10 through 15 requires correction by adding 6 even though it has no fifth-bit carry.

Rubrics:

- 1 mark for each correct answer; for part (d), the explanation is required.

[Bonus: 2] Design a circuit that takes a 4-bit binary number as input and outputs 1 if the number is odd and 0 if it is even. Mark deduction for a complicated solution.

No solution is provided for the bonus question, as instructed.